Algebra 1 | Name Answer Key___________
Unit 4 | Date___________________
Lesson 11 Homework | Period_________________

1. Write the equation of the line that passes through $(-2,1)$ and is perpendicular to $y = 2x - 3$ in point-slope form. Slope $= -\frac{1}{2}$

$y - 1 = \frac{-1}{2}(x - -2)$

$y - 1 = -\frac{1}{2}(x + 2)$

2. Write the equation in slope-intercept form of the line that is perpendicular to $x = 4$ and passes through point $(4, -5)$.

$y = -5$

3. Write the equation of the line that contains point $(-2,2)$ and is perpendicular to $x - 2y = 5$ in standard form. Slope $= -2$

$-x \quad -x$

$\frac{-2y}{-2} = \frac{5-x}{-2}$

$y = \frac{1}{2}x - \frac{5}{2}$

$y - 2 = -2(x + 2)$

$y - 2 = -2x - 4$

$+2x + 2 \quad + 2x + 2$

$2x + y = -2$

4. Write the equation of the line that is perpendicular to $y + 2 = 0$ and passes through point $(3, -3)$.

$x = 3$

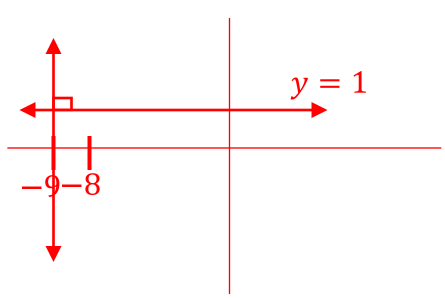
5. Line B passes through $(11, 13)$ and is perpendicular Line C which has an undefined slope. What is the slope of line B?

$m = 0$

6. Write the equation of the line that is perpendicular to the line that passes through the points $(0,1)$ and $(-8,1)$ and passes through point $(-9, 12)$.

$\frac{1-1}{-8-0} = \frac{0}{-8} = 0 =$ slope

Equation of line perpendicular → $x = -9$



$y - 1 = 0(x - 0)$

$\perp$ slope → undefined $\frac{8}{0}$

7. Write the equation of a line that is perpendicular to $y = x$ and contains point $(-6, 6)$.

Slope $= -1$

$6 = (-1)(-6) + b$

$6 = 6 + b$

$-6 \quad -6$

$b = 0$

$y = -x$

8. Write the equation of the line, in all three forms, that passes through point $(5, 15)$ and is perpendicular to $y + 6 = \frac{5}{8}(x - 24)$. Slope $= -\frac{8}{5}$

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 15 = -\frac{8}{5}(x - 5)$	$15 = 5\left(\frac{-8}{5}\right) + b$ $15 = -8 + b$ $\underline{+8 \quad +8}$ $23 = b$ $y = -\frac{8}{5}x + 23$	$y - 15 = \frac{-8}{5}x + 8$ $+\frac{-8}{5}x + 15$ $+\frac{-8}{5}x + 15$ $\left(\frac{8}{5}x + y = 23\right)5$ $8x + 5y = 115$

9. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(0, -9)$ and $(54, 0)$ and passes through point $(1, 9)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{0-(-9)}{54-0} = \frac{9}{54} \rightarrow \frac{-54}{9} = -6$ $y - 9 = -6(x - 1)$	$9 = -6(1) + b$ $9 = -6 + b$ $\underline{+6 \quad +6}$ $15 = b$ $y = -6x + 15$	$y - 9 = -6x + 6$ $+6x + 9 \quad +6x + 9$ $6x + y = 15$

10. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(-14, 22)$ and $(2, -2)$ and passes through point $(0, 4)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{-2-22}{2+14} = -\frac{24}{16} \rightarrow \frac{16}{24} = \frac{2}{3}$ $y - 4 = \frac{2}{3}(x - 0)$	$4 = 0\left(\frac{2}{3}\right) + b$ $4 = b$ $y = \frac{2}{3}x + 4$	$\left(y - 4 = \frac{2}{3}x\right)3$ $3y - 12 = 2x$ $-2x + 12 \quad -2x + 12$ $-2x + 3y = 12$ or $2x - 3y = -12$